Date:

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Soundex

Soundex is an algorithm used to find words that sound simillar, not necessarily spelled the same way. For example, "chebyshev" and "tchebyscheff" can be considered simillar by Soundex.

- Soundex Algorithm Steps:
  1. Keep the first letter
     Write down the first letter of the word.
  2. Replace specific letters with "0"
     Change these letters to '0': A, E, I, O, U, H, W, Y.
  3. Convert the remaining letters to digits.
     - B, F, P, V → 1
     - C, G, J, K, Q, S, X, Z → 2
     - D, T → 3
     - L → 4
     - M, N → 5
     - R → 6
  4. Remove Consecutive Duplicate Digits
     Remove one digit from any pair of the same number that appears together.
  5. Remove zeros and Pad with zeros
     Delete all the zeros. Then, add zeros at the end to make a 4-charater code (1 letter + 3 digits).

- Task: Create a soundex from the words "Yudhy" and "Yudi"
  1. Word 1: "Yudhy"
     - Step 1:
       Keep the first letter: 'Y.'
     - Step 2 & 3:
       Convert the rest:

Date:

+ U → 0

+ D → 3

+ H → 0

+ Y → 0

This gives : Y6300

- Step 4

Remove consecutive duplicate digits.

"00" becomes one "0", so it becomes: Y030

- Step 5

Remove all zeros : We get Y3.

Pad with zeros to get four characters : Y300

Soundex code for "Yudhy" = Y300

2. Word 2 : "Yudi"

- Step 1 :

Keep the first letter : Y.

- Step 2 & 3 :

Convert the rest :

+ U → 0

+ D → 3

+ I → 0

This gives : Y030

- Step 4 :

There are no consecutive duplicate digits to remove.

It remains : Y030

- Step 5 :

Remove zeros : We get Y3.

Pad with zeros to get four charaters : Y300

Soundex code for "Yudi" = Y300

- Conclusion

Both "Yudhy" and "Yudi" result in the code Y300. This shows that the Soundex algorithm considers these words Phonetically similar even if they're spelled dif.